Exploratory Data Analysis in Power BI

Simple EDA in Power BI

What we need to do

In this exercise, we need to **independently explore** a given dataset. We will have to **make decisions** at each step based on some prompts and **continuously reflect** on our findings.

We need to use a dataset that contains information about countries and their Sustainable Development Goals (SDG) scores and ranks.
Download it here: UN_SDG_dashboard_data.csv

Download this train to add insight and findings based on the prompts. **Review and provide feedback** on each other's insights.

An overview of the exercise

The goal of this exercise is to **encourage exploration**, **critical thinking**, and the development of **analytical skills** rather than focusing on obtaining a "correct" answer. The open-ended nature of the questions is designed to **promote curiosity**, **experimentation**, and **discussion** among peers.

01. Initial exploration
02. Data understanding
03. Comparative analysis
04. Grouping and aggregation
05. Correlation exploration
06. Goal-specific analysis
07. Iterative exploration
08. Synthesis and insights
09. Reflection and conclusion

Throughout this exercise, see if you can identify whether you've applied univariate or multivariate and graphical or non-graphical EDA methods.

Initial exploration

Initial exploration: To familiarise ourselves with the dataset by examining its structure, column names, and initial content, and to formulate questions or hypotheses based on the available information.

01. **Import** the data.

- What are the columns in the dataset? What information do they contain?
- Are there any immediate questions or hypotheses we have about the data based on the column names?

Data understanding

Data understanding: To gain an initial understanding of the distribution of a selected variable, using visualisations to identify patterns, outliers, and potential insights.

02. Choose one variable

(for example, `SDG Index Score`) and **create a visualisation** that can help us to **understand its distribution.**

- What insights can we gather from this visualisation?
- How might the distribution of this variable impact the analysis?

Comparative analysis

Comparative analysis: To explore relationships between two variables through visualisations, enabling the identification of patterns, trends, or correlations that may inform further analysis.

03. Select two variables

and create a **comparative visualisation** (for example, scatter plot, bar chart).

- What does this comparison reveal about the relationship between these variables?
- Are there any patterns or outliers that catch our attention?

Grouping and aggregation

Grouping and aggregation: To group data by a categorical variable (for example, region) and visualise aggregated values, facilitating the identification of variations or similarities among groups.

04. Group the data by a categorical variable

(for example, `Regions used for the SDR`) and **visualise the aggregated values**.

- What differences or similarities do we observe among groups?
- How might this grouping influence the interpretation of the data?

Correlation exploration

Correlation exploration: To calculate and interpret the correlation coefficient between two numeric variables, aiding in understanding the strength and direction of their relationship.

05. Calculate the correlation coefficient

for any two numeric variables.

- What does the correlation coefficient tell us about the relationship between these variables?
- How might this correlation impact the analysis or interpretation of the dataset?

Goal-specific analysis

Goal-specific analysis: To examine the distribution of scores for a specific SDG goal, using visualisations to identify variations or trends and relate them to the broader dataset.

06. Explore the distribution of scores

for one of the SDG goals (for example, Goal 1) using a suitable visualisation.

- What variations or trends do you notice in the scores for this specific goal?
- How might these insights contribute to the broader understanding of the dataset?

Iterative exploration

Iterative exploration: To encourage iterative exploration by selecting a variable of interest and using a different visualisation type, fostering a more comprehensive understanding of the data.

07. Choose another variable

and explore it further with a different type of visualisation.

- How does this new perspective enhance your understanding of the data?
- Does it prompt any additional questions or areas of interest?

Synthesis and insights

Synthesis and insights: To synthesise key insights gained from the exploratory analysis, encouraging reflection on unexpected findings and prompting further questions or areas of interest.

08. Summarise the key insights

gained from the exploratory analysis by building a report or dashboard.

- Are there any unexpected findings or patterns that emerged during the exploration?
- What further questions or analyses would we propose based on our initial discoveries?

Reflection and conclusion

Reflection and conclusion: To reflect on the overall exploration process, considering how initial questions evolved and what challenges were encountered, and to conclude with insights gained from the analysis.

09. Reflect on the overall exploration process.

- Did your initial questions evolve as you explored the data?
- What challenges or uncertainties did you encounter during the analysis?
- How did this exploratory analysis contribute to a better understanding of the dataset?